

Pseudomonas aeruginosa: Isolation and Characterization

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Suggested Citation

Monawer, A.T. (2024). *Pseudomonas aeruginosa*: Isolation and Characterization. *European Journal of Theoretical and Applied Sciences*, 2(3), 446-464.
DOI: [10.59324/ejtas.2024.2\(3\).35](https://doi.org/10.59324/ejtas.2024.2(3).35)

Abstract:

A total samples were collected from patients who attended the main hospitals in Duhok city during the period from 8th May 2022 to 28th February 2023. Bacterial identification, and antimicrobial susceptibility were tested using the traditional methods and confirmed by VITEK 2 compact system. Fifty selected *P. aeruginosa* isolates were molecularly detected by conventional PCR assay using specific primers and sequencing of *aac(6')-Ib* gene. For the prevalence of *P. aeruginosa*, 160 isolates were isolated from 670

samples including, burn 48(7.17%), ear 39(5.82%), wound 25(3.73%), urine 27(4.03%), and sputum 21(3.13%). Regarding antibiotic-resistant pattern, resistance was noticed to tobramycin, gentamicin, and amikacin showed resistance of 64.4%, 57.5%, and 51.9%, respectively. On molecular study, 50 isolates from 160 were selected for PCR assay. Selected isolates were confirmed by PCR assay that 46/50 (92%) were positive after *aac(6')-Ib* gene with size 472 bp amplification, and have got clear bands on agarose gel 1% and electrophoresis, 10 PCR positive samples were selected to know the sequencing of *aac(6')-Ib* genes. Moreover, all sequences were submitted to National Center for Biotechnology Information blast, recorded into Genbank, and got accession numbers for *aac(6')-Ib* (OQ538204, OQ538205, OQ538206, OQ538207, OQ538208, OQ538209, OQ538210, OQ538211, OQ538212, and OQ538213). The *aac(6')-Ib* samples were identical (99-100%) to the references sequence.

Keywords: *aac(6')-Ib*, antibiotics, *P. aeruginosa*, PCR, VITEK.

Introduction

Pseudomonas aeruginosa is one of the most important and opportunistic pathogens that causes a high rate of mortality and mortality in hospitalized patients with compromised immune systems (Aghamollaei et al., 2019). It is one of the most important hospital-acquired infection in burnt patients (Al-Daraghi and Al-Badrwi, 2020). *P. aeruginosa* has a tendency to form biofilms, which are compound bacterial groups that stick to a variety of surfaces together with plastics, medical transplant materials, and tissue. They are very difficult to destroy (Al-Daraghi and Al-Badrwi, 2020).

The frequent occurrence of drug resistance and persistent colonization on surfaces makes *P. aeruginosa* particularly difficult to treat and eradicate. Multiple intrinsic and acquired resistant mechanisms are exploited by *P. aeruginosa*, including the modification of drug targets, attenuation of membrane permeability, and formation of biofilms, which, collectively, contribute to its distinctly low antibiotic susceptibility (Taylor et al., 2014). *P. aeruginosa* has been classified as an ESKAPE (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* species) pathogen; one of the six highly antibiotic-resistant bacteria and known for their increasing bacterial virulence (Chen et

al., 2018), *P. aeruginosa* is designated by the World Health Organization (WHO) as a top antibiotic-resistant “priority pathogen” for which new antibiotics are critically required (WHO, 2017). *P. aeruginosa* is the most non-fermenting organism isolated and identified from the clinical specimens, which is resistant to various antibacterial drugs (Valentine et al., 2020). Infections caused by bacterial pathogens with multidrug-resistant (MDR) are challenging and difficult to treat (Mirzaei et al., 2020). *P. aeruginosa* is resistant to aminoglycosides and carbapenems; currently, the most effective treatment options are being increasingly reported (Tahmasebi et al., 2020).

The aminoglycosides (tobramycin, gentamicin, and amikacin) are commonly used to treat hospital-acquired infections caused by *P. aeruginosa*. These infections generally require treatment with a combination of antimicrobials in order to achieve a greater bactericidal effect and reduce the levels of resistance (Strateva and Yordanov, 2009). The use of a combination of antimicrobials, however, is associated with resistance mediated by aminoglycoside modifying enzymes (AME). Four of these enzymes, encoded by *aac* (6’)-Ib, *aac* (6’)-II, (2’)-I, and *aph* (3’)-VI, are of particular significance because they are among the most common modifying enzymes present in *P. aeruginosa*, and their substrates are the most important antipseudomonal aminoglycosides (Bertinellys et al., 2016). The spread of these enzymes can occur through genetic elements exchanged between the same or different taxa and is favored by the selective pressures present in a hospital environment where there is constant use of antimicrobial compounds. This has led to bacterial resistance which is becoming an increasing threat to public health (Bertinellys et al., 2016).

P. aeruginosa has been described as a ‘phenomenon of bacterial resistance’ so the present study aim to isolate and identify *P. aeruginosa* bacterium.

Materials

Place and Period of the Study

The study was carried out in the Microbiology Laboratory, Department of Basic Science, College of Nursing, University of Duhok from 8th May 2022 till 28th February 2023. The study was approved by the Ethics Committee /Scientific Research Division /Directorate of Planning/Duhok General Health Directorate/Ministry of Health/Kurdistan Regional Government/Iraq [Reference number: 22062021-6-8].

Bacterial Isolation

This study included 670 samples that were collected from different clinical sources (sputum, ear swab, urine, wound, and burn) in the main hospitals in Duhok city, Kurdistan region, Iraq; patients were involved both sexes and different age groups, samples were transported to the Microbiology Laboratory, within 1-2 hrs for culturing, and bacteriological identification. All samples were first cultured using sterile cotton swabs in sterile vials containing Nutrient broth and Amies transport medium swabs that were incubated at 37°C/24 hrs for microbiological identifications. To isolate *P. aeruginosa* each sample was first identified according to general cultural characteristics of colonies (color, shape, and size) grown on Nutrient agar, Blood agar, and MacConkey agar that were incubated at 37°C for 24hrs for further tests.

Identification of *Pseudomonas aeruginosa*

Bacterial identification was done by using Gram stain, biochemical tests (catalase test, oxidase test, IMViC test, and Triple Sugar Iron test), Cetrimide agar, and confirmation test was done by using automated VITEK 2 compact system.

Antimicrobial Susceptibility Test (Kirby Bauer Disk Diffusion Test)

For all identified *P. aeruginosa* isolates, a disc diffusion test for antibiotic susceptibility testing was performed as recommended by the Clinical Laboratory Standards Institute CLSI (CLSI, 2020) and the European Committee on Antimicrobial Susceptibility Testing EUCAST

(EUCAST, 2021) using tobramycin 10 µg (TOB), amikacin 30 µg (AK), gentamicin antibiotics. Sterile nutrient broths were prepared and well-isolated colonies of the bacterium growing on a nutrient agar plate were inoculated into it. The broths were incubated at 37°C until the culture equaled 0.5 McFarland standard. A McFarland 0.5 turbidity standard corresponds to an inoculum of 1.5×10^8 CFU/ml. If the suspension was too light, more colonies were added; if it was too heavy, then it was diluted with the sterile normal saline. The prepared suspension was used within 15min. A sterile cotton swab was dipped into the inoculum and wiped evenly across the surface of Muller Hinton agar plate (Okezie et al., 2021). The inoculated plates were placed at room temperature to allow absorption of excess moisture, then the antibiotic discs were placed firmly on the inoculated plates with forceps to ensure contact with the agar, then incubated at 37°C for 24 hrs. After incubation, the diameters of the inhibition zones were measured in mm. and compared with that of the NCCLS (CLSI, 2020).

Molecular identification (Extraction of DNA)

For DNA extraction, the boiling method was performed, by taking two to three fresh colonies from a bacterial plate and mixing them in 800 µl of deionized water. Then placed in a water bath at 100°C for 10 min followed by centrifugation at 1200 rpm for 2 minutes using a micro centrifuge (Mini Spin Plus, Eppendorf). Crude

lysate containing DNA was stored at -20°C for use over a long period (Jain and Barman, 2017).

Determination of DNA Concentration and Purity

The concentration and purity of the DNA of each isolate were determined using a Nanodrop machine (Molecular Department, General Central Laboratory, General Health Directorate in Duhok). Confirmation of the quantity and purity of the DNA was calculated by the Nanodrop 2000 spectrophotometer (Thermo Scientific Germany). The purity of the DNA was judged based on an optical density ratio of 1.7-1.8 and a concentration between 10-50 nanog /µl (Jain and Barman, 2017).

Conventional Polymerase Chain Reaction (PCR)

The DNA of *P. aeruginosa* isolates was targeted for the aac (6')-lb gene using the primer listed in (Table 1). A reaction mixture (20 µl) contained 3 µl of DNA, 1 µl of each primer, 10 µl of Master Mix, and 5 µl of nuclease-free water [18]. The conventional PCR machine is set up with many cycles and different temperatures for each step, including step one, Initial denaturation. Followed by step two (denaturation, annealing, and extension), and finally an extra extension. PCR program for aac (6')-lb gene is summarized in (Table 2), then the PCR products were analyzed using 1.5% agarose gel electrophoresis, and the ethidium bromide-stained bands in the gel were visualized using a Gel imaging system (Jain and Barman, 2017).

Table 1. Primer Sequences Used for PCR

Gene	Primer sequence (5'-3')	Product size (bp)	Reference
aac (6')-lb -F	5- TTGCGATGCTCTATGAGTGGCTA-3	472	Asghar and Ahmed, 2018
aac (6')-lb -R	5- CTCGAATGCCTGGCGTGT'T-3		

Table 2: Steps of Thermal Cycling, (PCR Program) for Gene aac (6')-lb

Steps	Temperature °C	Time m:s	Cycle
Initial denaturation	94	04:00	1
Denaturation	94	01:00	35
Annealing	50	01:00	
Extension	72	01:30	
Final extension	72	05:00	1
Hold	10	10:00	

DNA Sequencing

After DNA extraction from *P. aeruginosa* isolates, ten bacterial isolates were selected to perform DNA sequencing by sending the PCR products of the ten samples with aac (6')-Ib primers to Macrogen biotechnology company (Korea), to know the sequence of the genes using ABI 3130 genetic analyzer ((ABI PRISM® BigDye™ Terminator Cycle Sequencing Kits (Applied Biosystem). The gene sequences for bacterial isolates were applied to the Basic Local Alignment Search Tool (BLAST) which is a searching tool that applies the sequence alignment method and is available at the National Center for Biotechnology Information

(NCBI) website, and the bacterial isolates were recorded into GenBank data.

Statistical Analysis

All data statistical analysis was done using SPSS version 20 (IBM USA).

Results

A total of 670 different clinical samples from (burn, wound, ear, urine, and sputum) were aseptically collected from patients. The samples were screened for isolation of *P. aeruginosa*. There were 160 isolates of *P. aeruginosa* (23.88%), while 510 (76.12%) showed no bacterial growth (Figure 1).

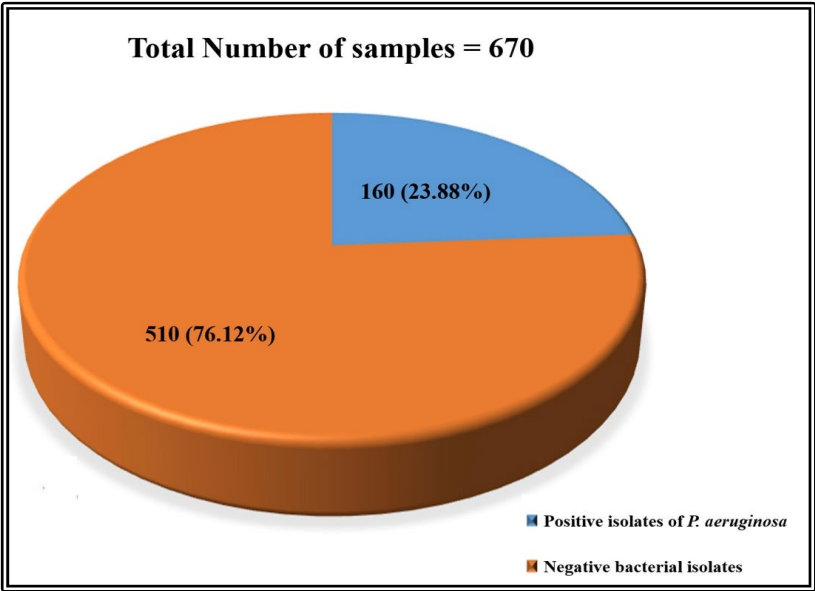


Figure 1. Number and Percentage of *P. aeruginosa* Isolates

The highest expression of *P. aeruginosa* was among the samples of burn and ear swabs (7.17% and 5.82%, respectively), followed by

urine (4.03%), and wound (3.73%), while the lowest rate was in sputum swabs (3.13 %) (Table 3).

Table 3. Isolates of *P. aeruginosa* from Different Clinical Samples

Sources	Number of samples (%)	Positive (%)	Negative (%)
Burn	120 (17.9%)	48 (7.17%)	72 (10.75%)
Ear	131 (19.6%)	39 (5.82%)	92 (13.73%)
Wound	142 (21.2%)	25 (3.73%)	117(17.46%)

Urine	220 (32.8%)	27 (4.03%)	193 (28.81%)
Sputum	57 (8.5%)	21 (3.13%)	36 (5.37%)
Total	670 (100%)	160 (23.88%)	510 (76.12%)

Table 4 showed the rates of antibiotic resistance in total *P. aeruginosa* isolates, the resistance was 64.4% for tobramycin. Gentamicin gave a

resistance of 57.5%. The least level of resistance was observed for amikacin 51.9%.

Table 4. Rates of Antibiotic Resistance in Total *P. aeruginosa* Isolates

Antibiotics	Symbol	Disc potency (µg)	No. (%)
Tobramycin	TOB	10	103 (64.4)
Gentamicin	CN	10	92 (57.5)
Amikacin	AK	30	83 (51.9)

For the distribution of *aac(6')-Ib* Genes, Out of the 160 *P. aeruginosa* isolates, 50 isolates were studied and selected for PCR. The *aac(6')-Ib*

gene was detected in 46/50 (92%) isolates (Figures 2).

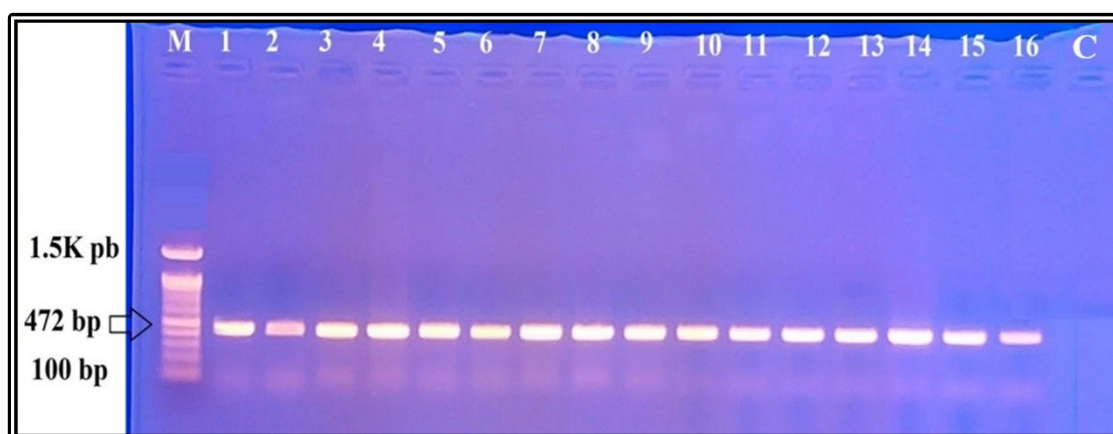


Figure 2. PCR product of the gene *aac(6')-Ib* in *P. aeruginosa* isolates. The amplification of DNA appears as a ladder-like pattern where, Lane (M) DNA marker (100 bp), Lane (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, and 16) with molecular weight 472 bp represent positive isolates, and Lane (C) indicate negative control without band.

DNA Sequencing Result

During this study, after molecular analysis ten bacterial isolates were sent to Macrogen Biotechnology Company (Korea) for DNA

sequencing, and then recorded into Genbank. Each bacterial isolate obtained a separate accession number for each gene (*blaOXA24* and *aac(6')-Ib*) (Table 5).

Table 5. The Accession Numbers of *P. aeruginosa* Bacterial Identification Depend on *blaOXA 24* and *aac(6')-Ib* Genes Submitted in the GenBank

Isolate No.	Species name	Accession number
		<i>aac(6')-Ib</i>
1	<i>P. aeruginosa</i>	OQ538204
2	<i>P. aeruginosa</i>	OQ538205

3	<i>P. aeruginosa</i>	OQ538206
4	<i>P. aeruginosa</i>	OQ538207
5	<i>P. aeruginosa</i>	OQ538208
6	<i>P. aeruginosa</i>	OQ538209
7	<i>P. aeruginosa</i>	OQ538210
8	<i>P. aeruginosa</i>	OQ538211
9	<i>P. aeruginosa</i>	OQ538212
10	<i>P. aeruginosa</i>	OQ538213

On the National Center for Biotechnology Information (NCBI) site, after typing the GenBank isolate accession number in the search field, the following information about aac (6')-Ib

for each bacterial isolate (<https://blast.ncbi.nlm.nih.gov/Blast>) (figures 3-12).

GenBank

Pseudomonas aeruginosa strain aac(6)-Ib-1 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds

GenBank: OQ538204.1

FASTA
Graphics

LOCUS
DEFINITION
ACCESSION
VERSION
KEYWORDS
SOURCE
ORGANISM
REFERENCE
AUTHORS
TITLE
JOURNAL
FEATURES
ORIGIN

OQ538204
403 bp
DNA
linear
BCT 24-APR-2023
Pseudomonas aeruginosa strain aac(6)-Ib-1 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds.
OQ538204
OQ538204.1
Pseudomonas aeruginosa
Pseudomonas aeruginosa
Bacteria; Pseudomonadota; Gammaproteobacteria; Pseudomonadales; Pseudomonadaceae; Pseudomonas.
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Monawer, A.T. and Abdulkahar, I.M.
Direct Submission
Submitted (26-FEB-2023) Nursing College-Dep. Basic Sciences, Duhok University, Sumil, Duhok, Duhok 38, Iraq
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Figure 3. aac(6')-Ib Gene Sample under the Accession Number (OQ538204) in (NCBI) (<https://blast.ncbi.nlm.nih.gov/Blast>) was Reported as *P. aeruginosa*

451

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Pseudomonas aeruginosa strain aac(6)-Ib-2 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds

GenBank: OQ538205.1

[FASTA](#) [Graphics](#)[Go to:](#)

LOCUS OQ538205 403 bp DNA linear BCT 24-APR-2023

DEFINITION Pseudomonas aeruginosa strain aac(6)-Ib-2 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds.

ACCESSION OQ538205

VERSION OQ538205.1

KEYWORDS .

SOURCE Pseudomonas aeruginosa

ORGANISM [Pseudomonas aeruginosa](#)
Bacteria; Pseudomonadota; Gammaproteobacteria; Pseudomonadales; Pseudomonadaceae; Pseudomonas.

REFERENCE 1 (bases 1 to 403)

AUTHORS Monawer, A.T. and Abdulkahar, I.M.

TITLE Direct Submission

JOURNAL Submitted (26-FEB-2023) Nursing College-Dep. Basic Sciences, Duhok University, Sumil, Duhok, Duhok 38, Iraq

FEATURES

source

Location/Qualifiers

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ORIGIN

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Figure 4. aac(6')-Ib Gene Sample under the Accession Number (OQ538205) in (NCBI) (<https://blast.ncbi.nlm.nih.gov/Blast>) was Reported as *P. aeruginosa*

Pseudomonas aeruginosa strain aac(6)-Ib-3 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds

GenBank: OQ538206.1

[FASTA](#) [Graphics](#)[Go to:](#) (v)

LOCUS OQ538206 403 bp DNA linear BCT 24-APR-2023

DEFINITION Pseudomonas aeruginosa strain aac(6)-Ib-3
fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase
AAC(6')-Ib (aac(6')-Ib) gene, partial cds.

ACCESSION OQ538206

VERSION OQ538206.1

KEYWORDS .

SOURCE Pseudomonas aeruginosa

ORGANISM [Pseudomonas aeruginosa](#)
Bacteria; Pseudomonadota; Gammaproteobacteria; Pseudomonadales;
Pseudomonadaceae; Pseudomonas.

REFERENCE 1 (bases 1 to 403)

AUTHORS Monawer, A.T. and Abdulkahar, I.M.

TITLE Direct Submission

JOURNAL Submitted (26-FEB-2023) Nursing College-Dep. Basic Sciences, Duhok
University, Sumil, Duhok, Duhok 38, Iraq

FEATURES

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ORIGIN

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Figure 5. aac(6')-Ib Gene Sample under the Accession Number (OQ538206) in (NCBI) (<https://blast.ncbi.nlm.nih.gov/Blast>) was Reported as *P. aeruginosa*

Pseudomonas aeruginosa strain aac(6)-Ib-4 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds

GenBank: OQ538207.1

[FASTA](#)
[Graphics](#)
Go to: 

LOCUS OQ538207 403 bp DNA linear BCT 24-APR-2023

DEFINITION Pseudomonas aeruginosa strain aac(6)-Ib-4 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds.

ACCESSION OQ538207

VERSION OQ538207.1

KEYWORDS .

SOURCE Pseudomonas aeruginosa

ORGANISM [Pseudomonas aeruginosa](#)
Bacteria; Pseudomonadota; Gammaproteobacteria; Pseudomonadales; Pseudomonadaceae; Pseudomonas.

REFERENCE 1 (bases 1 to 403)

AUTHORS Monawer, A.T. and Abdulkahar, I.M.

TITLE Direct Submission

JOURNAL Submitted (26-FEB-2023) Nursing College-Dep. Basic Sciences, Duhok University, Sumil, Duhok, Duhok 38, Iraq

FEATURES

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ORIGIN

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Figure 6. aac(6')-Ib Gene Sample under the Accession Number (OQ538207) in (NCBI) (<https://blast.ncbi.nlm.nih.gov/Blast>) was Reported as *P. aeruginosa*

Pseudomonas aeruginosa strain aac(6)-Ib-5 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds

GenBank: OQ538208.1

[FASTA](#) [Graphics](#)

LOCUS OQ538208 403 bp DNA linear BCT 24-APR-2023

DEFINITION Pseudomonas aeruginosa strain aac(6)-Ib-5
fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase
AAC(6')-Ib (aac(6')-Ib) gene, partial cds.

ACCESSION OQ538208

VERSION OQ538208.1

KEYWORDS .

SOURCE Pseudomonas aeruginosa

ORGANISM [Pseudomonas aeruginosa](#)
Bacteria; Pseudomonadota; Gammaproteobacteria; Pseudomonadales;
Pseudomonadaceae; Pseudomonas.

REFERENCE 1 (bases 1 to 403)

AUTHORS Monawer, A.T. and Abdulkahar, I.M.

TITLE Direct Submission

JOURNAL Submitted (26-FEB-2023) Nursing College-Dep. Basic Sciences, Duhok
University, Sumil, Duhok, Duhok 38, Iraq

FEATURES

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Figure 7. aac(6')-Ib Gene Sample under the Accession Number (OQ538208) in (NCBI) (<https://blast.ncbi.nlm.nih.gov/Blast>) was Reported as *P. aeruginosa*

Pseudomonas aeruginosa strain aac(6)-Ib-6 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds

GenBank: OQ538209.1

[FASTA](#) [Graphics](#)[Go to:](#) (v)

LOCUS OQ538209 403 bp DNA linear BCT 24-APR-2023

DEFINITION Pseudomonas aeruginosa strain aac(6)-Ib-6 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds.

ACCESSION OQ538209

VERSION OQ538209.1

KEYWORDS .

SOURCE Pseudomonas aeruginosa

ORGANISM [Pseudomonas aeruginosa](#)
Bacteria; Pseudomonadota; Gammaproteobacteria; Pseudomonadales; Pseudomonadaceae; Pseudomonas.

REFERENCE 1 (bases 1 to 403)

AUTHORS Monawer, A.T. and Abdulkahar, I.M.

TITLE Direct Submission

JOURNAL Submitted (26-FEB-2023) Nursing College-Dep. Basic Sciences, Duhok University, Sumil, Duhok, Duhok 38, Iraq

FEATURES

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Figure 8. aac(6')-Ib Gene Sample under the Accession Number (OQ538209) in (NCBI) (<https://blast.ncbi.nlm.nih.gov/Blast>) was Reported as *P. aeruginosa*

Pseudomonas aeruginosa strain aac(6)-Ib-7 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds

GenBank: OQ538210.1

[FASTA](#)
[Graphics](#)
[Go to:](#)

LOCUS OQ538210 403 bp DNA linear BCT 24-APR-2023

DEFINITION Pseudomonas aeruginosa strain aac(6)-Ib-7 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds.

ACCESSION OQ538210

VERSION OQ538210.1

KEYWORDS .

SOURCE Pseudomonas aeruginosa

ORGANISM [Pseudomonas aeruginosa](#)
Bacteria; Pseudomonadota; Gammaproteobacteria; Pseudomonadales; Pseudomonadaceae; Pseudomonas.

REFERENCE 1 (bases 1 to 403)

AUTHORS Monawer, A.T. and Abdulkahar, I.M.

TITLE Direct Submission

JOURNAL Submitted (26-FEB-2023) Nursing College-Dep. Basic Sciences, Duhok University, Sumil, Duhok, Duhok 38, Iraq

FEATURES

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Figure 9. aac(6')-Ib Gene Sample under the Accession Number (OQ538210) in (NCBI) (<https://blast.ncbi.nlm.nih.gov/Blast>) was Reported as *P. aeruginosa*

Pseudomonas aeruginosa strain aac(6)-Ib-8 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds

GenBank: OQ538211.1

[FASTA](#)
[Graphics](#)

Go to:

LOCUS OQ538211 403 bp DNA linear BCT 24-APR-2023

DEFINITION Pseudomonas aeruginosa strain aac(6)-Ib-8
fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase
AAC(6')-Ib (aac(6')-Ib) gene, partial cds.

ACCESSION OQ538211

VERSION OQ538211.1

KEYWORDS

SOURCE Pseudomonas aeruginosa

ORGANISM [Pseudomonas aeruginosa](#)

Bacteria; Pseudomonadota; Gammaproteobacteria; Pseudomonadales;
Pseudomonadaceae; Pseudomonas.

REFERENCE 1 (bases 1 to 403)

AUTHORS Monawer, A.T. and Abdulkahar, I.M.

TITLE Direct Submission

JOURNAL Submitted (26-FEB-2023) Nursing College-Dep. Basic Sciences, Duhok
University, Sumil, Duhok, Duhok 38, Iraq

FEATURES Location/Qualifiers

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Figure 10. aac(6')-Ib Gene Sample under the Accession Number (OQ538211) in (NCBI) (<https://blast.ncbi.nlm.nih.gov/Blast>) was Reported as *P. aeruginosa*

Pseudomonas aeruginosa strain aac(6)-Ib-9 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds

GenBank: OQ538212.1

[FASTA](#)
[Graphics](#)
[Go to:](#) ☐

LOCUS OQ538212 403 bp DNA linear BCT 24-APR-2023

DEFINITION Pseudomonas aeruginosa strain aac(6)-Ib-9 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds.

ACCESSION OQ538212

VERSION OQ538212.1

KEYWORDS .

SOURCE Pseudomonas aeruginosa

ORGANISM [Pseudomonas aeruginosa](#)
Bacteria; Pseudomonadota; Gammaproteobacteria; Pseudomonadales; Pseudomonadaceae; Pseudomonas.

REFERENCE 1 (bases 1 to 403)

AUTHORS Monawer, A.T. and Abdulkahar, I.M.

TITLE Direct Submission

JOURNAL Submitted (26-FEB-2023) Nursing College-Dep. Basic Sciences, Duhok University, Sumil, Duhok, Duhok 38, Iraq

FEATURES

source Location/Qualifiers

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Figure 11. aac(6')-Ib Gene Sample under the Accession Number (OQ538212) in (NCBI) (<https://blast.ncbi.nlm.nih.gov/Blast>) was Reported as *P. aeruginosa*

Pseudomonas aeruginosa strain aac(6)-Ib-10 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds

GenBank: OQ538213.1

[FASTA](#) [Graphics](#)[Go to:](#)

LOCUS OQ538213 403 bp DNA linear BCT 24-APR-2023

DEFINITION Pseudomonas aeruginosa strain aac(6)-Ib-10 fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase AAC(6')-Ib (aac(6')-Ib) gene, partial cds.

ACCESSION OQ538213

VERSION OQ538213.1

KEYWORDS .

SOURCE Pseudomonas aeruginosa

ORGANISM [Pseudomonas aeruginosa](#)
Bacteria; Pseudomonadota; Gammaproteobacteria; Pseudomonadales; Pseudomonadaceae; Pseudomonas.

REFERENCE 1 (bases 1 to 403)

AUTHORS Monawer, A.T. and Abdulkahar, I.M.

TITLE Direct Submission

JOURNAL Submitted (26-FEB-2023) Nursing College-Dep. Basic Sciences, Duhok University, Sumil, Duhok, Duhok 38, Iraq

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Figure 12. aac(6')-Ib Gene Sample under the Accession Number (OQ538213) in (NCBI) (<https://blast.ncbi.nlm.nih.gov/Blast>) was Reported as *P. aeruginosa*

The DNA sequencing of *aac(6')*-Ib gene that amplified by using PCR that was submitted in GenBank. The BLAST analysis was showed that

the *aac(6')*-Ib gene shared 99.5% to 100% homology with the sequences of *P. aeruginosa*. (Table 6).

Table 6. *Pseudomonas aeruginosa* Isolates and Genbank Accession Number of *aac(6')*-Ib Gene

Isolate No.	Species name	Accession No.	Similarity %	References Accession No.	Country
1	<i>P. aeruginosa</i>	OQ538204	100	MF772756.1	Kenya
2	<i>P. aeruginosa</i>	OQ538205	99.75	NG_051711	USA
3	<i>P. aeruginosa</i>	OQ538206	99.75	CP101885	Argentina
4	<i>P. aeruginosa</i>	OQ538207	99.75	EU161636	Hungary
5	<i>P. aeruginosa</i>	OQ538208	99.50	LR134330	UK
6	<i>P. aeruginosa</i>	OQ538209	99.50	CP029088	USA
7	<i>P. aeruginosa</i>	OQ538210	99.50	MF687204	Italy
8	<i>P. aeruginosa</i>	OQ538211	99.50	KY574887	Thailand
9	<i>P. aeruginosa</i>	OQ538212			
10	<i>P. aeruginosa</i>	OQ538213			

Discussion

Isolation and Identification of *Pseudomonas aeruginosa*

In the current study, out of 670 clinical samples that were obtained and collected from patients, 160 (23.88%) isolates of *P. aeruginosa* were detected and identified, this result is similar to a previous study in Spain done by Causapé and colleagues, who indicated that 79/341 (23.2%) of *P. aeruginosa* were identified (Causapé *et al.*, 2017). In addition, Al-Saffar and Jarallah, found that 20 (25%) isolates from 80 samples collected from different clinical and environmental sources were identified as *P. aeruginosa* in AL-Hilla Teaching Hospital/ Babylon/ Iraq (Al-Saffar and Jarallah, 2019). A study in Baghdad detected 40 (26.7%) *P. aeruginosa* out of 150 samples (Al-Daraghi and Al- Badrwi, 2020). In the other direction, a study done by reported AL-Fridawy *et al.*, that *P. aeruginosa* isolates were 54 (36%) out of 150 (AL-Fridawy *et al.*, 2020). This rate was higher than the current study and this may be due to differences with regard to the sources of samples and sites of isolation. In earlier research done by other researchers, pointed out that the rate of occurrence of *P. aeruginosa* (51.50%) is the highest among other bacteria isolated from clinical samples included in the study (Hussein *et al.*, 2018).

Distribution of *Pseudomonas aeruginosa* among Different Clinical Samples

In the current study, the frequencies of *P. aeruginosa* isolates were detected at most from burn 48/120 (7.17%), ear samples 39/131(5, 82%), urine 27/220 (4.03%), and followed by wound 25/142 (3.73%), whereas the lowest rate was among sputum samples 21/57(3.13%). A study conducted at Minia University Hospital/ Egypt showed different rates which were urine (15/50), sputum (27/87), ear discharges (32/83), wound exudates (35/85), and stool (16/45) (Ghanem *et al.*, 2023). *P. aeruginosa* isolates were isolated in Turkey, among them 36 (18%) sputum, 34 (17%) urine, and 10 (5%) wound swabs (Cayci, *et al.*, 2022). Other studies showed different rates which were in contrast to this result in all different samples. For example, in Jeddah / Saudi Arabia, the majority (32.5%) of isolated *P. aeruginosa* were from urine, (18.2%) wounds, (13.6%) ear swabs, and (13%) sputum (Abdulrahman *et al.*, 2023). Other researchers have pointed out that the highest rate was in the ear (27.5%), then wound (12.5%), and burn (5%) (Al-Daraghi and Al-Badrwi, 2020).

Antimicrobial Susceptibility Patterns

Data from the present study showed aminoglycosides-resistance was as follows:

tobramycin (64.4%), gentamicin (57.5%), and amikacin (51.9%). These findings were in agreement with a previous study by Raytaker *et al.*, (2017) that showed a resistance rate to gentamycin (57.8%), resistance to tobramycin (64.4%) was slightly higher than detected in the study of Nasr El-din, *et al.*, that showed (60%) resistance to the antibiotic, while in the same study, amikacin resistance was (56%) which was higher than that of the current study (Nasr El-din *et al.*, 2022). On the contrary, data from the present study showed a difference from those obtained previously by Azizoglu *et al.*, (2021) who reported that tobramycin, gentamicin, and amikacin showed resistance rates (44%, 46%, and 40%), respectively, Ghanem *et al.*, (2023) reported that tobramycin (80.7%), gentamicin (69.1%), and amikacin (41.7%) on bacterial isolates, a result which disagrees with the results of this study. Resistance of clinical isolates to aminoglycoside antibiotics was found to vary with a specific drug, the microorganism, its mechanism of resistance, the geographic area, and many other factors (Mohamed *et al.*, 2019).

Molecular Detection of *Pseudomonas aeruginosa* isolates for aac(6')-Ib Gene

Regarding aac (6')-Ib gene, in the current study, our results showed that the frequency rates of aac (6')-Ib gene among the collected *P. aeruginosa* isolates distributed in different hospitalized patients was 46/50 (92%). These findings were compatible with Azimi and colleagues, who reported that the most frequently detected aac (6')-Ib gene was in (95.1%) of *P. aeruginosa* isolates (Azimi *et al.*, 2022). Also, this was highlighted by many other studies which were in concordance with our data, that showed aac(6')-Ib gene was the most frequently detected in *P. aeruginosa* isolates conducted in different countries, such as Venezuela, Iran, and Egypt (Teixeira *et al.*, 2016; Beigverdi *et al.*, 2020; Elmesiry *et al.*, 2022). On the other hand, other studies showed different rates which were in contrast to our data. For example, in Makkah / Saudi Arabia, the frequent rate of aac (6')-Ib was (6.1%, 4/65) (Asghar and Ahmed, 2018).

Conclusion

From the results of the present study, it was concluded that the majority of *P. aeruginosa* isolates were collected from burn, followed by ear, urine, wound, sputum swabs, and all *P. aeruginosa* isolates expressed good resistant to tobramycin, followed by gentamicin and amikacin. Moreover highly prevalence of aac(6')-Ib resistance genes among *P. aeruginosa* isolates.

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